

6.7800 Project Part 2

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1 Things I tried

1. Iterate through all positions assuming it is the breakpoint, running the decode algorithm on the first and second segments for each iteration. This is actually massively slow, as if you try to decrease iterations the accuracy drops off to the point of being pointless.
2. Instead of iterating, I implemented binary search, where you call decode on each half but with less iterations for efficiency, and whichever half has the worse decode rate is determined to contain the breakpoint. Once you reach this end, run the whole MCMC algorithm again but with the full number of iterations. This is still slow.
3. Still do binary search, but instead look at the bigram distribution and determine where it changes the most.